SPECTRA: A Traceability Ontology for the 3GPP RAN Standardization Process

Sihyeon Choi* and Junho Lee

System LSI Business, Device Solutions Division,
Samsung Electronics, South Korea
{shyun12.choi, junho515.lee}@samsung.com
* *First and corresponding author.*

Abstract. Standardization documents are produced through a dense, interlinked, and continuously revised workflow of technical contributions, working-group decisions, and change requests. Yet this standardization process is rarely captured as a queryable knowledge graph in the Semantic Web, leaving traceability tasks (recovering the contribution lineage behind a clause, mapping cross-group Liaison-Statement chains, aggregating per-company contribution patterns) to manual archive search. We present **SPECTRA** (*SPECification TRAcing*), an OWL ontology of the 3GPP Radio Access Network (RAN) standardization process. The schema is organized around four concerns common to standardization workflows — contributions, resolutions, specifications, and cross-group communication — and was developed iteratively through cumulative competency-question regression. We evidence schema-level reuse by instantiating the RAN1-designed schema across all five RAN Working Groups, which required no additional classes. The release accompanies *SpectraCQ v1.0*, a 137-question competency-question and Cypher query suite released as a separately citable benchmark. SPECTRA-authored components (ontology, SHACL shapes, documentation, queries, examples, validation scripts, and the sanitized parsing pipeline) are released under CC-BY 4.0; the bundled per-WG body-text knowledge graphs are 3GPP-derived literals redistributed with explicit 3GPP attribution.

Keywords: Ontology engineering · OWL · 3GPP · Standardization process · Document traceability · Competency questions · Semantic Web

1 Introduction

Standards-developing organizations (SDOs) such as the 3rd Generation Partnership Project (3GPP), the Internet Engineering Task Force (IETF), and the Institute of Electrical and Electronics Engineers (IEEE) produce evolving document corpora. The lifecycle of these documents—from initial contributions through working-group (WG) resolutions to normative specification text—shapes which technologies become global infrastructure. The Semantic Web has extensively modeled provenance [10] and related publication or software artifacts, but the SDO standardization *process itself* remains under-modeled, even though SDOs produce artifacts that govern interoperability across the telecom, networking, and semiconductor industries. We focus on a large and practically important under-modeled SDO setting: 3GPP, the global SDO defining 4G Long-Term Evolution (LTE) and 5G NR (New Radio), and coordinating ongoing 6G standardization. The Radio Access Network (RAN) Technical Specification Group (TSG) alone encompasses five Working Groups (RAN1–RAN5), each responsible for a distinct layer of the radio interface, from physical-layer procedures to conformance testing.

3GPP specifications are the normative reference that commercial 5G implementations and conformance tests must align with. Yet the corpus spans hundreds of normative specifications and evolves through a process that produces hundreds of thousands of Working Group documents. Acquiring even partial working knowledge typically requires substantial domain expertise and prolonged immersion. This creates barriers along three axes: *cross-WG*, where a RAN1 decision can propagate to the sibling RAN WGs (RAN2–RAN5) through Liaison Statements (LSs) that are difficult to track end-to-end; *cross-organization*, where each company has internal context mainly for its own contributions; and *industry–academia*, where researchers see the published specifications but not the meeting-level decisions that shaped them. The traceability path TDoc → Resolution → CR → TS Section → Spec, which helps recover the document-level lineage behind a clause and locate where it resides, currently exists as implicit institutional knowledge and is typically reconstructed by hand — error-prone and time-consuming.

To the best of our knowledge, after surveying the published Semantic Web and 3GPP-domain literature (ISWC/ESWC proceedings 2014–2025, the Linked Open Vocabularies registry, and 3GPP/GSMA dataset releases), no publicly available OWL ontology specifically targets the 3GPP RAN standardization document lifecycle. Recent 3GPP-domain resources — TSpec-LLM [11] for retrieval-augmented access to specification text and *telecom-kg-rel19* [23], a GraphML entity–relation KG with provenance and text chunks built from Release-19 specification text — model the *content* of specifications but are not designed to directly represent the standardization *process* that produced them. Examples of such process-level questions include: which TDocs led to a clause, which CRs responded to or were linked from a Liaison Statement, or which Working Assumptions were promoted to Agreements.

In this paper, we present **SPECTRA** (*SPECification TRAcing*), an OWL resource for modeling the types, relationships, and lifecycle of 3GPP RAN standardization documents. The schema's four-concern structure (contributions, resolutions, specifications, and cross-group communication) appears to map onto sibling SDOs whose document lifecycles share the same shape (contributions are tabled, working-group resolutions decide their fate, change requests amend the normative text, and cross-group communication propagates impact); empirical validation outside 3GPP RAN is left to future work. The contributions are:

1. **Traceability tasks SPECTRA supports** — a practically important family of cross-document queries that are otherwise reconstructed by hand: recovering the document-level lineage of a clause's current text (S1, multi-hop trace), mapping cross-WG Liaison-Statement chains (S2), aggregating Release-scoped CR analytics (S3), Working-Assumption promotion lineage *captured by the schema for future data-side population* (S4), propagating TR-to-TS impact (S5), per-company contribution profiles (S6), and Work-Item-scoped lineage (S7). Each task is shown in §6 with the schema fragment it exercises; S1–S3 and S5–S7 are populated in the released/deployed snapshots, while S4 is currently a schema-level pattern.

2. **Public release package** — SPECTRA-authored components (the OWL ontology with 26 classes, 53 object properties, 81 data properties; reuses Dublin Core, DCTERMS, FOAF; optional PROV-O alignment; SHACL shapes, PyLODE documentation, representative Cypher/SPARQL queries, synthetic and metadata-only examples, a cross-WG metadata-only process KG over RAN1–RAN5, and the sanitized parsing pipeline source) under CC-BY 4.0, alongside per-WG body-text KGs as 3GPP-derived literals retained with explicit 3GPP attribution following [11,23]; release details and licensing tiers in Section 7.

3. **Cross-WG and reproducibility-aware validation evidence** — the schema is instantiated across all five RAN Working Groups (Table 6); applying the RAN1-designed schema to RAN2–RAN5 required *no additional classes beyond the RAN1-designed schema*, only three schema-level extension candidates (one OP and two DPs: RAN3 `summaryId`, RAN5 `discussionText`; Table 8); per-WG class coverage ranges 76.9–100% (§6.4), reflecting released-snapshot scope rather than schema gaps. The release ships the schema, validation JSONs, the cross-WG process KG, per-WG body-text KGs (RAN1–RAN5, with 3GPP attribution), and the sanitized parsing pipeline; reviewers can re-derive the artifact-integrity claims directly via `verify_release.py`, while full 137-CQ Cypher correctness depends on the operational KG and is recorded in `validation/cq_results.json`. Raw Neo4j database dumps and pre-computed embedding vectors are excluded (large binary artifacts; regenerable from the bundled body-text KGs and pipeline source).

4. ***SpectraCQ v1.0* benchmark; release-time cumulative CQ regression discipline as construction method** — the 137-CQ dataset (NL question + executable Cypher + RAN1 verdict) is released as separately citable *SpectraCQ v1.0* (CC-BY 4.0) at `cqs/spectra_cq_v1.0/` for SDO-process ontology benchmark reuse. SPECTRA was developed through five incremental phases (P1–P5; 25/34/45/15/18 CQs totalling 137; phase mapping in §5) with cumulative regression checks at each transition that surfaced and resolved three QA cases before release; this construction method complements established CQ-driven methodologies (LOT [16], SAMOD [13], NeOn [20]) rather than replacing them.

2 Related Work

2.1 Ontologies for Domain Knowledge Management

Ontologies have been widely adopted as the semantic backbone for domain-specific knowledge management systems. In cultural heritage, the Polifonia Ontology Network [2] provides 15 modular ontologies for musical heritage, developed through eXtreme Design [18] with 361 competency questions from 10 pilots. The Hydrogen Ontology (HOLY) [1] models the hydrogen market with over 100 classes covering organisations, projects, components, products, applications, markets, and indicators, developed via the Linked Open Terms (LOT) methodology with stakeholder competency questions from the Atlant-H project. In industry, Cao et al. [4] describe a knowledge graph for the quality management of Bosch products as an example of industrial KG deployment. For technical standards, Grangel-González and Vidal [6] use the Standards Ontology (STO) as the schema for an Industry 4.0 Knowledge Graph covering more than 200 industrial standards, but their focus is on *inter-standard* relationships rather than the *standardization-process document lifecycle* (sections, change requests, meeting decisions, liaison statements) that is central to our work. In telecommunications, NORIA-O [21] targets anomaly detection and incident management in ICT systems rather than the standardization process itself. Closer in spirit, formal document lifecycles have been modelled outside telecommunications by UNDO [14] (UN documents, generic Workflow vocabulary) and the IETF RFC ontology [15] (published-RFC bibliographic metadata: `obsoletes/updates/Errata`); neither targets an SDO's working-group deliberation events (meeting-level resolutions, cross-WG liaison routing, working-assumption-to-agreement promotions), which are the SPECTRA core. A recent effort in the 3GPP domain is *telecom-kg-rel19* [23], a Hugging Face dataset distributing a GraphML entity–relation KG with provenance and text chunks derived from Release-19 specification *text* for retrieval-augmented LLM reasoning. Unlike that resource, SPECTRA is an OWL ontology of the standardization process and document lifecycle: it models TDoc, Resolution, CR, LS, TR, Section, and related process entities rather than concept structures embedded in specification text. Table 1 positions SPECTRA against the closest comparable resources along the axes most relevant to the ISWC Resource Track.

2.2 Ontology Engineering Methodologies

Ontology 101 [12] popularized a practical ontology-development workflow, and competency questions (CQs) have long served as the primary requirements mechanism [7]. LOT [16] extends this for industrial settings with FAIR compliance. The NeOn methodology [20] provides scenario-based pathways for building ontology networks, including reuse of existing ontological and non-ontological resources. SAMOD [13] draws on Test-Driven Development principles. SPECTRA's five-phase protocol (§5) is complementary rather than a replacement: the accumulated CQ suite is re-executed at each internal phase transition so earlier-phase competencies are revisited each time the schema grows. This prac-

Table 1. Comparison with related Semantic Web resources along Resource-Track-relevant axes (each work's published abstract / availability statement). *Process* = explicit domain process / lifecycle model; *multi-hop trace* = backward traceability over $\geq$ 3 class types; *CQ-grounded* = explicit CQ set; *cross-WG/subdomain* = released schema validated across $\geq$ 2 sibling WGs or subdomains in the stated domain. For SPECTRA this is RAN1–RAN5 of the RAN TSG (one schema, five sibling WGs); for Polifonia, ontology-network modules covering distinct musical subdomains.

Resource	Domain	Process	Multi-hop trace	CQ-grounded	Cross-WG/ subdomain
Polifonia [2]	cultural heritage	limited	n/a	yes (361 CQs)	yes (15 modules)
HOLY [1]	hydrogen market	limited	n/a	yes (LOT/CQs)	n/a
Bosch QM [4]	industrial QM	limited	n/a	not reported	n/a
NORIA-O [21]	ICT incidents	yes (incident)	yes (incident RCA)	yes (NOC/SOC)	n/a
Grangel-G. & Vidal [6]	Industry 4.0 stds	limited	n/a	not reported	n/a
UNDO [14]	UN documents	yes (workflow)	limited	not reported	n/a
RFC ontology [15]	RFC metadata	limited	limited (updates)	not reported	n/a
TSpec-LLM [11]	3GPP text	n/a	n/a	n/a	n/a
telecom-kg-rel19 [23]	3GPP Rel-19 text	n/a	n/a	n/a	n/a
SPECTRA (ours)	3GPP RAN proc.	yes	yes (5-class, join-aware)	yes (137 CQs)	yes (5 WGs)

Axes chosen to surface differences in process modelling, multi-hop traceability, and CQ-grounded design relevant to the Resource Track rubric. "n/a" means the axis is outside the cited resource's stated design focus, not a claim that the resource is incapable of it. Under these axes, SPECTRA is the only listed resource focused on 3GPP RAN process modelling with document-lifecycle traceability and cross-WG instantiation evidence.

tice surfaced three regression cases (§5) that single-phase validation would not have detected before release.

In terms of CQ scale, the Polifonia Ontology Network [2] releases a 361-CQ dataset (*PolifoniaCQ*) collected across the network's modules, and Wisniewski et al. [22] translated 234 CQs into 131 SPARQL-OWL queries. SPECTRA contributes 137 CQs applied cumulatively to a single ontology spanning five document-lifecycle areas: TDoc metadata, meeting resolutions, TS structure, CR documents, and technical reports.

3 3GPP Standardization Background

3.1 Document Types and Lifecycle

SPECTRA models the main interlinked artifact families in the 3GPP RAN document lifecycle (Figure 1): TDocs (with CR and LS subtypes), Resolutions, Specifications/Sections, and Technical Reports. **TDoc** is the primary contribution unit submitted as a temporary document to a meeting (e.g., R1-2400001); **CR** (Change Request modifying normative text) and **LS** (Liaison Statement carrying communication to other 3GPP groups or to external technical bodies) are subtypes of TDoc per 3GPP's own usage. **Resolution** captures meeting decisions (Agreement/Conclusion/Working Assumption) and is modelled as a first-class SPECTRA superclass so that per-meeting decisions are queryable regardless of subtype. **TS** (Technical Specification, e.g., TS 38.211) carries the normative text in hierarchical Sections; **TR** (Technical Report — informative study/feasibility material) links to the Specs it influences via reified `TRImpact`.

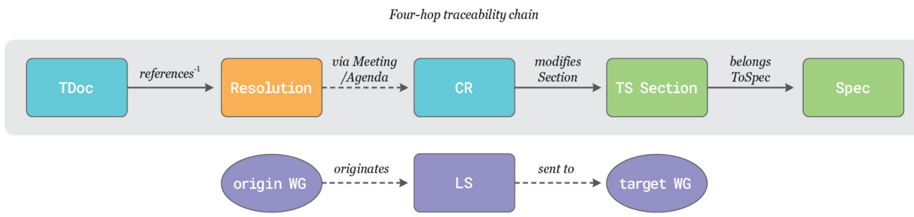

Fig. 1. Document lifecycle in 3GPP RAN. Top: reader-facing traceability path TDoc → Resolution → CR → Section → Spec. Formal OPs are `references` (Resolution → Tdoc, drawn reversed as `references`$^{-1}$), `modifiesSection`, `belongsToSpec`; the dashed Resolution–CR step is an indirect meeting/agenda/CR-pack join rather than a direct OP. Bottom: LS-based cross-WG communication via `originatedFrom`/`sentTo`. The sixth document type *TR* reaches the spine through the reified `TRImpact` relation (Figure 2).

3.2 Working Group Structure

The RAN TSG is divided into five WGs whose responsibilities follow 3GPP's official scope statements: **RAN1** (Radio Layer 1, e.g., TS 38.211–214); **RAN2** (Radio Layer 2 protocols and Radio Resource Control, e.g., TS 38.331/321); **RAN3** (NG-RAN architecture and related interfaces, e.g., TS 38.401/423); **RAN4** (Radio Performance and Protocol Aspects, e.g., TS 38.101/104); **RAN5** (Mobile terminal conformance testing, e.g., TS 38.521/533). Inter-group dependencies are mediated by Liaison Statements addressed to other 3GPP WGs and to external technical bodies; RAN WG meetings convene on an approximately two-month cadence (varying 6 weeks to 3 months, with e-meetings during 2020–2022) and may involve thousands of TDocs.

Generalization beyond 3GPP. Table 2 sketches an analogy to IETF and IEEE because the four-concern shape (contributions, resolutions, specifications, and cross-group communication) appears to be common across these SDOs; the IETF/IEEE rows are document-class analogues, and a real instantiation with its own elicited CQs is left to future work.

Table 2. Four-concern analogy across three SDOs. The lifecycle shape (contribution → resolution → amendment → specification/report artifact) appears comparable; vocabulary and process specifics differ.

SDO	Contribution	Resolution	Amendment	Specification artifact
3GPP (this paper)	TDoc, LS	Agreement / Conclusion / Working Assumption	Change Request (CR)	TS (normative); TR (informative)
IETF (analogy)	Internet-Draft	WG consensus / IESG approval	bis-draft / RFC errata [9]	RFC
IEEE 802 (analogy)	Submission, Comment	Letter Ballot, motion	PAR amendment, sponsor ballot	IEEE Std

3.3 Concepts an Ontology Must Capture

Three features shape the ontology: **scale** (10^4–10^5 TDocs per WG; cf. Table 6); **document lineage** (TDoc→Resolution→CR→Section→Spec demands first-class types and directed links per document); and **cross-group dependencies** (Liaison Statements as directed artefacts, not free text). These requirements inform the four lifecycle concerns and supporting class clusters described in §4.

4 SPECTRA Ontology Design

4.1 Competency Question Analysis

SPECTRA is grounded in 137 competency questions collected across five development phases by practicing 3GPP RAN delegates within the authors' organization (external-SME validation deferred, see §8). Every property exercised by the 137 CQs maps to a field publicly available on 3GPP TDoc cover sheets, chairman-note resolutions, or per-meeting `TDOC_List.xlsx`—the dataset is structurally grounded in externally verifiable 3GPP records rather than internal-only metadata. The **137-CQ dataset (*SpectraCQ v1.0*)** retains the verbatim public 3GPP company identifiers from `TDOC_List.xlsx` where they appear (14 of 137 CQs). The dataset — English text, Cypher query specifications validated on the internal KG, phase, category, schema area, and RAN1 per-CQ verdict — is shipped at `cqs/spectra_cq_v1.0/` under CC-BY 4.0; the per-CQ PASS verdict is carried into the public `validation/cq_results.json` reproduced by `tests/verify_release.py`, and a representative subset runs against the bundled synthetic and metadata-only mini examples.

CQs are distributed across the development phases as: P1 TDoc metadata (25, e.g. "What TDocs did Huawei submit?"), P2 Meeting resolutions (34, "What agreements on Agenda Item Z?"), P3 TS structure (45, "Which CRs modified section S?"), P4 CR documents (15, "What is the reason for change of CR C?"), P5 Technical reports (18, "Which TRs have impacted TS 38.211?"), totalling 137.

4.2 Core Schema

Based on the CQ analysis, we designed SPECTRA following the Ontology 101 methodology [12]. The ontology is encoded in OWL 2 [8] and comprises 26 classes, 53 object properties (OPs), and 81 data properties (DPs). These classes operationalize the four lifecycle concerns introduced in §1 (contributions, resolutions, specifications, cross-group communication) plus organizational and auxiliary-content clusters needed for joins and extracted evidence (Figure 2).

Document: `Tdoc` with four subclasses (`CR`, `LS`, `Summary`, `Session Notes`); `CR` → `Spec` via `modifies` and `CR` → `Section` via `modifiesSection`; `LS` → `WorkingGroup` via `sentTo`/`originatedFrom`. **Resolution**: subclasses `Agreement`, `Conclusion`, `WorkingAssumption` (with `promotedTo` linking a `WorkingAssumption` to a later `Agreement`, often after cross-WG or cross-meeting dependencies are resolved). **Specification**: `Spec` (TS) and `Section` forming a tree via `hasSubSection`/`parentSection` (irreflexive, asymmetric); `TechnicalReport` links to influenced Specs through a reified `TR`

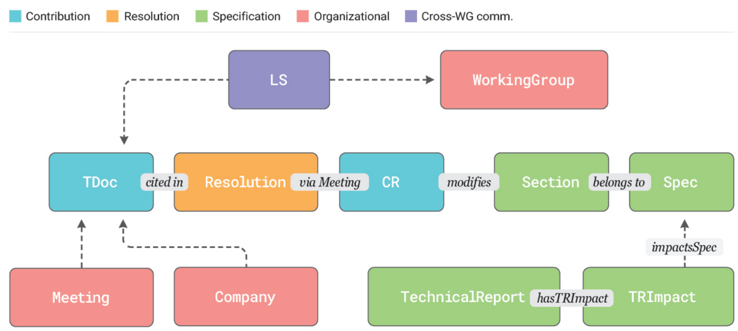

Fig. 2. SPECTRA core schema (reader-facing labels). **Central row**: traceability spine `Tdoc` $\rightarrow$ `Resolution` $\rightarrow$ `CR` $\rightarrow$ `Section` $\rightarrow$ `Spec`; formal OPs are `references` (drawn reversed), `modifiesSection`, and `belongsToSpec`. The dashed Resolution $\rightarrow$ CR step is a `Meeting/AgendaItem/CRPack` join rather than a direct OP. **Dashed grey arrows**: secondary OPs (`sentTo`, `presentedAt`, `submittedBy` — reader-facing) and LS $\sqsubseteq$ Tdoc subclass edge; the TR $\rightarrow$ TRImpact $\rightarrow$ Spec reified chain is shown explicitly. Section/TR sub-tree edges (irreflexive `hasSubSection/parentSection`, `referencesSection/referencesTR`, and inverses) are documented in `spectra.ttl`.

`Impact` so a single TR can affect multiple Specs with per-impact metadata. **Organizational**: `Meeting`, `Company` ($\sqsubseteq$ `foaf:Organization`), `Contact` ($\sqsubseteq$ `foaf:Person`), `WorkItem`, `AgendaItem`, `Release`, `WorkingGroup`. **Auxiliary content**: `Figure`, `Table`, `Chart`, `TSTable`, `TSFigure` capture structured content extracted from contributions and specifications for downstream traceability tools (figure-to-section linkage, table-to-CR provenance) even though they sit off the main lifecycle spine; instance counts are reported in Table 6.

Key Design Decisions. **D1:** Document types are OWL subclasses (not property values) so that type-specific properties can be declared per subclass. **D2:** 20 object properties carry `owl:FunctionalProperty` for an *at-most-one* contract (e.g., a Section `belongsToSpec` at most one Spec; each `TRImpact` points to at most one impacted Spec, while a single TR may have multiple `TRImpact` nodes); the loader enforces these at ingest time. Re-tableable links (`presentedAt`) and links that are legitimately multi-valued, implicit, or source-dependent (`modifies`, `originatedFrom/sentTo`) are deliberately not declared functional — see the data-quality notes in `validation/ran1_relation_integrity.json`. **D3:** The spine (Figure 2) is realized by three direct OPs (`references`, `modifiesSection`, `belongsToSpec`); the Resolution $\rightarrow$ CR step is an implementation-level join combining `madeAt` (Resolution $\rightarrow$ Meeting), `presentedAt` (CR $\rightarrow$ Meeting), and the shared `AgendaItem` (via `resolutionBelongsTo` from Resolution and `belongsTo` from CR/Tdoc); for TSG CR-pack approval flows, `belongsToCRPack` (CR $\rightarrow$ CRPack) augments the join via the literal `tsgMeeting` attribute on CRPack—no single OP, by design, so each declared property keeps single, verifiable semantics. **D4:** TR-to-Spec impact is reified as `TRImpact` rather than a direct OP because a single TR study typically affects multiple Sections of multiple Specs with a per-impact `impactType` (one of *NewTS, NewSection, ExtendSection, NoChange*; range `xsd:string` in OWL with the closed value set enforced by SHACL `sh:in` in `shapes/spectra-core.shacl.ttl`) carried alongside the impacted section/clause; a direct OP would lose that per-impact context. **D5:** An *optional* PROV-O alignment ships as 6 `rdfs:subClassOf` axioms (`Resolution` $\sqsubseteq$ `prov:Activity`; `Tdoc` $\sqsubseteq$ `prov:Entity`; `Company` $\sqsubseteq$ `prov:Agent`, `prov:Organization`; `Contact` $\sqsubseteq$ `prov:Agent`, `prov:Person`); we deliberately avoid `owl:imports` so consumers without a PROV reasoner pay no cost while reusers can opt in for cross-domain provenance queries.

4.3 Reuse of Existing Vocabularies

SPECTRA reuses established Semantic Web vocabularies where they naturally apply (Table 3). Domain-specific classes are introduced only where no suitable external term exists.

Table 3. Reuse of external vocabularies in SPECTRA, grouped by usage area.

Vocabulary	Slot	Terms used
Dublin Core [5]	Metadata	`dc:title`, `dc:description`, `dc:creator`, `dc:date`, `dc:rights`
DCTERMS	Licensing	`dcterms:license`
FOAF [3]	Agent class	`foaf:Person` ($\sqsupseteq$ `Contact`), `foaf:Organization` ($\sqsupseteq$ `Company`)
OWL 2	Axioms	`Functional/InverseFunctional/Irreflexive/Asymmetric Property`, `inverseOf`
RDFS	Hierarchy	`subClassOf, label, comment`
PROV-O [10] (optional)	Provenance	6 `rdfs:subClassOf` alignment axioms (Resolution$\sqsubseteq$Activity; Tdoc$\sqsubseteq$Entity; Company/Contact$\sqsubseteq$Agent; Company$\sqsubseteq$Organization; Contact$\sqsubseteq$Person)

We considered but did *not* adopt **ORG** (Organization/Unit/Membership beyond CQ needs), **SKOS** (hierarchical term taxonomies rather than typed entities), or **DCAT/VoID** (dataset description). The optional PROV-O alignment ships as 6 `rdfs:subClassOf` axioms in the OWL file but `owl:imports` is not asserted, so consumers can opt in without pulling the full PROV-O TBox; SPECTRA itself supplies the 3GPP-specific roles (TDoc, CR, LS, Agreement, Working Assumption, AgendaItem, CRPack, TS Section).

4.4 Constraints and Axioms

SPECTRA includes 20 functional + 2 inverse-functional (`hasCR`, `hasTRImpact`) properties, 15 inverse pairs, 6 irreflexive (section hierarchy, intra-section + TR cross-references), and 2 asymmetric (section hierarchy) declarations. SHACL shapes (`shapes/spectra-core.shacl.ttl`) capture cardinality and range portably—e.g., a `spectra:Tdoc` is required to carry exactly one `tdocNumber` (`sh:minCount 1; sh:maxCount 1`) and a `Section` carries exactly one `belongsToSpec` link (`sh:minCount 1; sh:maxCount 1`). Lifecycle links with multi-valued, implicit, or source-dependent behavior (`presentedAt` under re-tabling, `originatedFrom/sentTo` for cross-WG LSs, `modifies` for paired CRs) are deliberately left without strict cardinality bounds; their data-quality conventions are documented in the data-quality notes in `validation/ran1_relation_integrity.json`.

5 Cumulative CQ Quality Assurance

This section describes the cumulative CQ regression practice that produced the released ontology — a QA discipline complementary to established CQ-driven methodologies (LOT, SAMOD, NeOn): each new phase must leave every previously validated competency intact, so that schema refinements are caught at the point of introduction rather than at release.

5.1 Five-Phase Progressive Construction

SPECTRA was developed through five incremental phases — **P1** TDoc cover-sheet metadata, **P2** meeting-level resolutions (Agreement / Conclusion / WorkingAssumption), **P3** TS section structure, **P4** CR documents and CR-pack semantics, **P5** TR impact on TS — each extending the ontology while re-executing the full CQ suite accumulated from prior phases. The schema element trajectory (classes / object properties / data properties) is 11/14/31 at P1+P2 → 17/23/52 at P3 → 24/43/78 at P4 → 26/51/87 at P5, with v1.0.0 ending at 26/53/81 after refactoring identified through cross-phase integrity verification (e.g., `hasRelatedTR` added for explicit `WorkItem`–TR linkage; unused Meeting attributes removed; net −6 DPs and +2 OPs, with classes and 137 CQs unchanged). Cumulative CQ counts at the four phase transitions: 59 → 104 → 119 → 137. Per-phase TTL snapshots, scripts, and the cumulative CQ verdict files are released under `release_package/`.

5.2 Quality-Assurance Cases

Three representative cases illustrate what single-phase validation would have missed: **Case 1** (P2→P3) harmonized loader-side Cypher relationship labels (`MADEAT` vs `MADE_AT`, both mapping to `madeAt`) — multiple Phase-1–2 CQs would have silently regressed. **Case 2** (P3 audit) patched meeting-ID referential-integrity gaps that broke `Tdoc`→`Meeting`→`Resolution` joins. **Case 3** (P3→P4) resolved three schema refinements: `releaseId` removed, `hasCR` declared inverse-functional, and `Decision` renamed to `Resolution`.

5.3 Cumulative Regression Verification

At each transition, all CQs from Phases 1–N are re-validated against the Phase-N schema over the internally instantiated KG; the released artifact ships the final-state RAN1 verdict (`validation/cq_results.json`, 137/137 PASS), and per-phase verdicts are tracked in the development process. The 137 CQs accumulated by Phase 5 all return non-empty results on RAN1 and were manually verified; subsequent transitions completed without further regressions beyond the QA cases above.

6 Evaluation

We organise the evaluation around five dimensions: (E1) requirements coverage by CQs; (E2) structural soundness; (E3) instantiation scale; (E4) cross-WG reuse; (E5) public reusability. Release-scope and licensing rationale is in §7.

6.1 E1 — Requirements coverage by CQs

All 137 CQs return non-empty results on RAN1 (137/137 PASS; per-phase 25/34/45/15/18); `verify_release.py` verifies the released verdict files publicly (not re-execution against the operational KG, threat acknowledged in §8). The 137 CQs exercise 20 of 26 classes and 31 of 53 object properties. The 6 unexercised classes are `LS` plus Chart/Figure/Table/Summary/SessionNotes; LS-typed TDocs are exercised through their `Tdoc` superclass via three Phase-1 CQs covering `originatedFrom/sentTo/replyTo/ccTo`, and full LS chains run via use scenario S2 over `examples/process_kg/` (25,586 distinct LSs across the RAN1–RAN5 union; per-WG breakdown in `examples/process_kg/`). `modifiesSection` is exercised by use scenarios S1/S6; `promotedTo` is schema-defined (S4) with deployment-side population planned (Table 9).

6.2 E2 — Structural quality

We report structural metrics of the released ontology (Table 4) and the findings of an automated pitfall scan.

Table 4. Ontology structural metrics. The reused-vocabulary count excludes the 6 optional PROV-O alignment axioms.

Metric	Value
Classes	26
Object Properties	53
Data Properties	81
Functional Properties	20
Inverse Functional Properties	2
Inverse Property Pairs	15
Irreflexive Properties	6
Asymmetric Properties	2
Core reused vocabulary terms	8 (5 DC + 1 DCTERMS + 2 FOAF)

OOPS! [17] findings (Table 5) are 0 Critical and four Important/Minor categories with documented design rationales (P34 follows from FOAF reuse; P10 reflects a deferred `owl:disjointWith` decision because no CQ relies on disjointness reasoning). PyLODE HTML and SHACL conformance reports ship in the release (§7).

Table 5. OOPS! pitfall scan summary; design rationale shown for each finding.

Code	Severity	#	Design rationale
P34	Important	2	FOAF `Person/Organization` reused without redeclaration
P10	Important	1	`owl:disjointWith` on Tdoc subclasses deferred (CQs do not rely on it)
P08	Minor	2	Annotations omitted on reused FOAF terms
P13	Minor	23	`owl:inverseOf` declared only on bidirectionally queried pairs

6.3 E3 — Large-scale instantiation evidence

To provide scale evidence from the operational instantiation, we instantiated SPECTRA against RAN1 data. The graph materializes 123,677 unique RAN1 TDocs from 60 meeting sessions spanning 5G NR development (RAN1#84–#123 numbering range, Feb 2016 through Rel-19; the 60 sessions include bis/e-meeting variants); per-class instance counts are in Table 6 and per-relation referential-integrity in Table 7. references/modifiesSection coverage below 100% reflects 3GPP citation granularity rather than schema gaps. Aggregate counts, metadata-only derivatives, and per-WG body-text dumps are all in v1.0.0 (validation/, examples/process_kg/, kg/per_wg/; §7).

Table 6. Per-class instance counts in the RAN1 instantiation (operational snapshot, April–May 2026; validation/ran1_instance_counts.json). WorkingGroup=116 covers RAN1–RAN5 plus other TSGs and external bodies cited via originatedFrom/sentTo on LS cover sheets. The released ran1_tdoc_metadata.ttl preserves 123,678 source records resolving to 123,677 distinct identifiers (one TDoc re-tabled at a later meeting). ‡ Chart captures embedded OpenXML chart objects (<c:chart>) in DOCX Final Reports, distinct from raster Figure (<a:blip>); virtually all authors paste raster plots, so this class is rare by data-source convention.

Class	#	Class	#	Class	#
Tdoc	123,677	Resolution	24,710	Section	1,003
CR	10,715	Agreement	21,058	TSTable	680
LS	6,343	Conclusion	2,703	TSFigure	13
Summary	3,468	WorkingAssump.	949	CRPack	501
SessionNotes	678	Meeting	60	TechnicalReport	26
Spec	83	Company	222	TRImpact	29
Release	13	Contact	1,000	Figure	120
WorkingGroup	116	WorkItem	423	Table	1,441
AgendaItem	6,439			Chart ‡	1

Table 7. RAN1 source-derived relation coverage (validation/ran1_relation_integrity.json). presentedAt/madeAt omitted as loader-enforced (100%); sub-100% references/modifiesSection reflects 3GPP citation conventions, not schema gaps.

Property	Coverage	Numerator / Denominator
submittedBy	99.36%	154,525 / 155,525
references	51.55%	19,454 / 37,738
modifiesSection	34.66%	4,769 / 13,759

6.4 E4 — Cross-WG generality of the schema

A schema designed for one WG can fail silently when applied to a sibling WG with different document conventions. We applied SPECTRA — originally designed from RAN1 CQs — to RAN2–RAN5 without redesign and measured how many *additional* schema elements each WG needed (Table 8).

The 26 classes cover every document-type distinction the per-WG loaders produced (**zero** additional classes); the only WG-specific extensions are one object property (REFERENCES_SPEC) and two data properties (RAN3 summaryId, RAN5 discussionText) — schema-level reuse evidence within the released RAN1–RAN5 snapshots of the RAN TSG, not a universal transferability claim.

Conversely, 6 of the 26 RAN1 classes are absent from at least one sibling WG's instantiation; per-WG coverage ranges from 76.9% (RAN5, conformance-testing snapshot lacks Agreement, Chart, Figure, SessionNotes, Summary, WorkingAssumption) to 100% (RAN1) over the 26-class baseline; the full per-WG matrix and absent-class lists are in validation/per_wg_class_coverage.json. These absences reflect released-snapshot and loader scope, not schema gaps; an

Table 8. Schema diff of RAN2–RAN5 instantiations against the RAN1-designed SPECTRA baseline. "WG-specific extensions" counts schema elements observed in a non-RAN1 WG graph beyond the original RAN1 baseline (union over RAN2–RAN5); they are reported as extension candidates rather than evidence of class-level redesign. Extension rate is computed against the corresponding released-SPECTRA element count (1 of 53 OPs, 2 of 81 DPs).

Schema element	In released SPECTRA	WG-specific extensions	Extension rate
Classes	26	0	0.0%
Object Properties	53	1	1.9%
Data Properties	81	2 †	2.5%

† excluding 20 loader-local properties (17 underscore-prefixed mirror slots and 3 internal scaffolding properties); full per-WG breakdown in `validation/cross_wg_schema_diff.json`.

OWL diff of `kg/per_wg_schema/RAN{1..5}-schema.ttl` against `spectra.ttl` reproduces the per-WG class coverage (OP/DP extensions in `cross_wg_schema_diff.json`). Use scenarios S2–S3 (Table 9) additionally cite task-level cross-WG cardinalities measured on the deployed RAN2–RAN5 KGs.

6.5 E5 — Reusability evidence via public artifacts

To illustrate what the ontology is for, Table 9 lists seven curated competency-question scenarios. Each pairs a natural-language question with the schema fragment it exercises and the role it plays in the standardization workflow; S2–S3 cite cardinalities measured on the deployed RAN1–RAN5 KGs.

Table 9. Use scenarios S1–S7 supported by SPECTRA. Schema fragments cite OWL property names from the released TTL.

Scenario	Question	Schema fragment / signal
S1 Multi-hop trace	TDoc lineage behind current text of TS 38.214 §5.1.3	`Section` ← `CR` → `Meeting` ← `Resolution` → `Tdoc`; uses `modifiesSection`, `presentedAt`, `madeAt`, `references`
S2 Cross-WG LS impact	RAN2 contributions in meeting M responding to RAN1 LSs	`originatedFrom`, `sentTo`, `replyIn`; deployed: 3,644 LSs RAN1→RAN2 (top recipient); 1,450 received in RAN2 KG (peak RAN2#118-e: 62 received from RAN1)
S3 Release CR analytics	CRs against Release 18 by Huawei	`targetRelease`, `modifies`, `submittedBy`; deployed: TS 38.300 has 2,039 RAN2 + 270 RAN3 CRs
S4 WA promotion lineage	WAs promoted to Agreements (*schema captures the pattern*; data-side population by enhanced ingest is future work)	`WorkingAssumption` $\xrightarrow{\texttt{promotedTo}}$ `Agreement`, joined on `madeAt`
S5 TR→TS impact	TS sections affected by TR 38.889 conclusions	reified `TRImpact` via `hasTRImpact`, `impactsSection` (1-TR-to-many-TS)
S6 Per-company profile	TS 38.331 sections modified by CRs from Samsung over 4 meetings	`submittedBy`, `presentedAt`, `modifiesSection`, `belongsToSpec`
S7 Work-Item lineage	TDocs/CRs associated with agreements under WI `NR_eMIMO-Core`	`relatedToWorkItem`, `madeAt`, `references`, `hasCR`

SpectraCQ v1.0 ships at `cqs/spectra_cq_v1.0/`; cross-WG reuse evidence is at the schema level (`validation/cross_wg_schema_diff.json`).

7 Availability, Sustainability, and FAIRness

The **v1.0.0 release** adopts a two-tier license boundary — SPECTRA-authored components under CC-BY 4.0 and 3GPP-derived literals retained with explicit 3GPP attribution (cf. [11,23]) — distributed across GitHub (SPECTRA-authored layer plus per-WG schema-instantiation TTLs) and Zenodo (per-WG body-text KGs `RAN{1..5}-body.ttl`, ∼0.9 GB; literals carry `dcterms:rightsHolder=3GPP`). `validation_manifest.md` maps each quantitative claim to its supporting file.

FAIR-R alignment.

F: persistent IRI `https://w3id.org/spectra` (registered via `perma-id/w3id.org`); Zenodo DOI `10.5281/zenodo.20034872` (minted for the body-text deposit); LOV submission planned.

A: the ontology, SHACL, SpectraCQ, queries, examples, schema-instantiation TTLs, pipeline, and validation files are downloadable from the public GitHub repository without authentication; the per-WG body-text TTLs are downloadable from the Zenodo deposit.

I: OWL 2 Turtle (RDFLib-loadable); SHACL Core (`shapes/spectra-core.shacl.ttl`) validated under pyshacl; reuses DC, DCTERMS, FOAF; optional PROV-O alignment.

R: CC-BY 4.0 for SPECTRA-authored components; bundled 3GPP-derived body-text literals retain explicit 3GPP attribution (Tier 2); deterministic `verify_release.py` reports the per-check status of every claim referenced from `validation/validation_manifest.md` and exits non-zero on failure.

Sustainability. SPECTRA is the semantic layer of a Samsung System LSI standards-engineering KG with incremental ingestion after each RAN WG meeting (per the ∼2-month cadence noted in §3) via the released pipeline (`pipeline/`); a Qdrant-based semantic search layer over body-text KGs is also deployed internally but is out of Resources Track scope for the public release. The ontology itself evolves alongside the corpus: schema versioning follows semver (v1.x backward-compatible; breaking changes increment major version after a deprecation notice), and version transitions, deprecations, and per-version migration notes are tracked in `CHANGELOG.md` on the public GitHub mirror, where the issue tracker also captures schema-extension requests from external reusers. All 137 SpectraCQ Cypher queries use only SPARQL-equivalent constructs (`validation/cypher_to_sparql_portability.json`; 6 representative SPARQL translations bundled).

8 Conclusion

We presented **SPECTRA**, a reusable OWL ontology for the 3GPP RAN standardization document lifecycle, grounded in 137 CQs (*SpectraCQ v1.0* benchmark) and exercised through five-phase cumulative QA. Cross-WG reuse: only three extension candidates (one OP, two DPs) on RAN2–RAN5. **Threats:** CQs were verified by eliciting engineers rather than an external panel; adoption is confined to the authors' organisation; cross-SDO transferability (IETF/IEEE) is design intent rather than evidence; the released RAN1 snapshot exercises 20 of 26 classes by direct CQ (LS plus five auxiliary classes are reached through superclass paths and use scenarios); `references`/`modifiesSection` coverage is bounded by 3GPP citation conventions (51.6%/34.7% on the RAN1 snapshot); use scenario S4 (Working-Assumption promotion lineage) is currently a schema-level pattern with data-side population deferred. **Outlook.** This release initiates a longer-running effort to ground SDO process tracing in a formal ontology; subsequent releases will fold the internally deployed semantic search layer into the public release, add PROV-O process OPs, extend coverage to TSG SA/CT, explore LLM-assisted ontology generation from CQs [19], and pursue external evaluation.

Resource Availability Statement. **Code, schema, and small artifacts (GitHub).** `https://github.com/spectra-ontology/spec-trace` (CC-BY 4.0 for SPECTRA-authored components) ships **(i)** the OWL ontology (`ontology/spectra.ttl`, 26 classes / 53 OPs / 81 DPs; pyLODE-rendered HTML in `docs/spectra.html`); **(ii)** SHACL shapes (`shapes/spectra-core.shacl.ttl`, 8 NodeShapes); **(iii)** *SpectraCQ v1.0* (`cqs/spectra_cq_v1.0/`, 137 NL questions + 137 Cypher queries; a static scan confirms all 137 are schema-level patterns directly translatable to SPARQL, see `validation/cypher_to_sparql_portability.json`; CQs reference public 3GPP company identifiers verbatim where present); **(iv)** per-WG schema-instantiation TTLs (`kg/per_wg_schema/RAN{1..5}-schema.ttl`) so reviewers can reproduce the cross-WG generality numbers by OWL diff against `spectra.ttl`; **(v)** `validation/` (per-claim manifest + 11 supporting JSONs) and `tests/verify_release.py` (deterministic verifier); **(vi)** the sanitized parsing pipeline (`pipeline/`); **(vii)** `supplement/` (TTL/SHACL excerpts, SPARQL listings, reproducibility checklist, regression cases, LLM-baseline utility pilot summary). **Per-WG body-text KGs (Zenodo).** `RAN{1..5}-body.ttl` (∼0.9 GB) under DOI `10.5281/zenodo.20034872`; 3GPP-derived literals with explicit 3GPP attribution. **Persistent URI:** `https://w3id.org/spectra`. **Acknowledgments.** We thank colleagues at the System LSI Business of Samsung Electronics for domain feedback.

Declaration of use of Generative AI

AI coding assistants (Claude Code, OpenAI Codex) were used for English-prose editing, code generation and refactoring across the SPECTRA parsing/validation pipeline, textual rendering of author-decided ontology and shape designs into Turtle, and review-pass cross-validation against the released artifacts. An additional LLM was used for natural-language-to-Cypher translation suggestions during CQ-design iterations.

Generative AI was *not* used for the substantive scientific content of the paper — the original problem framing, ontology design decisions, and competency-question elicitation — which is author-driven. The authors take full responsibility for all content.

References

1. Ascencion Arevalo, K.M., Neunsinger, C., Zimmermann, R., Blum, R., Weakly, K.: HOLY: An ontology covering the hydrogen market. In: The Semantic Web – ISWC 2023, Resources Track. Springer (2023). https://doi.org/10.1007/978-3-031-47243-5_19
2. de Berardinis, J., Carriero, V.A., Jain, N., Lazzari, N., Meroño-Peñuela, A., Poltronieri, A., Presutti, V.: The Polifonia Ontology Network: Building a semantic backbone for musical heritage. In: The Semantic Web – ISWC 2023. pp. 302–322. Springer (2023). https://doi.org/10.1007/978-3-031-47243-5_17
3. Brickley, D., Miller, L.: FOAF Vocabulary Specification 0.99. http://xmlns.com/foaf/spec/ (2014)
4. Cao, Q., Grangel-González, I., Du, L.: A knowledge graph-based approach for the quality management of Bosch products. In: ISWC 2023 Posters, Demos, and Industry Tracks. CEUR Workshop Proceedings, vol. 3632 (2023), https://ceur-ws.org/Vol-3632/ISWC2023_paper_507.pdf
5. DCMI Usage Board: DCMI Metadata Terms. https://www.dublincore.org/specifications/dublin-core/dcmi-terms/ (2020)
6. Grangel-González, I., Vidal, M.E.: Analyzing a knowledge graph of industry 4.0 standards. In: Companion Proceedings of the Web Conference 2021. pp. 16–25. ACM (2021). https://doi.org/10.1145/3442442.3458600
7. Gruninger, M., Fox, M.S.: The role of competency questions in enterprise engineering. In: Rolstadås, A. (ed.) Benchmarking—Theory and Practice. pp. 22–31. Springer/IFIP (1995)
8. Hitzler, P., Krötzsch, M., Parsia, B., Patel-Schneider, P.F., Rudolph, S.: OWL 2 Web Ontology Language Primer (Second Edition). Tech. rep., W3C Recommendation (2012)
9. Kolkman, O., Bradner, S., Turner, S.: Characterization of Proposed Standards. IETF RFC 7127, https://datatracker.ietf.org/doc/html/rfc7127 (2014)
10. Lebo, T., Sahoo, S., McGuinness, D.: PROV-O: The PROV ontology. W3C Recommendation, https://www.w3.org/TR/prov-o/ (2013)
11. Nikbakht, R., Benzaghta, M., Geraci, G.: TSpec-LLM: An open-source dataset for LLM understanding of 3GPP specifications. In: 2024 IEEE Globecom Workshops (GC Wkshps). pp. 1–6 (2024). https://doi.org/10.1109/GCWkshp64532.2024.11101012
12. Noy, N.F., McGuinness, D.L.: Ontology development 101: A guide to creating your first ontology. Tech. Rep. KSL-01-05, Stanford Knowledge Systems Laboratory, Stanford University (2001)
13. Peroni, S.: A simplified agile methodology for ontology development. In: OWL: Experiences and Directions – Reasoner Evaluation (OWLED 2016). Springer (2016)
14. Peroni, S., Palmirani, M., Vitali, F.: UNDO: The United Nations System Document Ontology. In: The Semantic Web – ISWC 2017. LNCS, vol. 10588, pp. 175–183. Springer (2017). https://doi.org/10.1007/978-3-319-68204-4_18
15. Petit-Huguenin, M.: An ontology for RFCs. IETF Internet-Draft, draft-petithuguenin-rfc-ontology-05, https://datatracker.ietf.org/doc/draft-petithuguenin-rfc-ontology/05/ (2024)

16. Poveda-Villalón, M., Fernández-Izquierdo, A., Fernández-López, M., García-Castro, R.: LOT: An industrial oriented ontology engineering framework. Engineering Applications of Artificial Intelligence **111**, 104755 (2022). https://doi.org/10.1016/j.engappai.2022.104755
17. Poveda-Villalón, M., Gómez-Pérez, A., Suárez-Figueroa, M.C.: OOPS! (OntOlogy Pitfall Scanner!): An on-line tool for ontology evaluation. International Journal on Semantic Web and Information Systems (IJSWIS) **10**(2), 7–34 (2014). https://doi.org/10.4018/ijswis.2014040102
18. Presutti, V., Daga, E., Gangemi, A., Blomqvist, E.: eXtreme Design with Content Ontology Design Patterns. In: Workshop on Ontology Patterns (WOP 2009), co-located with ISWC 2009. CEUR-WS (2009)
19. Saeedizade, M.J., Blomqvist, E.: Navigating ontology development with large language models. In: The Semantic Web – ESWC 2024. pp. 143–161. Springer (2024). https://doi.org/10.1007/978-3-031-60626-7_8
20. Suárez-Figueroa, M.C., Gómez-Pérez, A., Fernández-López, M.: The NeOn Methodology for Ontology Engineering. In: Ontology Engineering in a Networked World, pp. 9–34. Springer (2012)
21. Tailhardat, L., Chabot, Y., Troncy, R.: NORIA-O: An ontology for anomaly detection and incident management in ICT systems. In: The Semantic Web – ESWC 2024. pp. 21–39. Springer (2024). https://doi.org/10.1007/978-3-031-60635-9_2
22. Wisniewski, D., Potoniec, J., Lawrynowicz, A., Keet, C.M.: Analysis of ontology competency questions and their formalizations in SPARQL-OWL. Journal of Web Semantics **59**, 100534 (2019). https://doi.org/10.1016/j.websem.2019.100534
23. Yang, Y., Bariah, L., Lu, Y., Debbah, M.: 3GPP release 19 telecom knowledge graph. Hugging Face dataset (KU-DF / GSMA Open Telco Assets Initiative), https://huggingface.co/datasets/GSMA/telecom-kg-rel19 (2026)